

Michael Khuri

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EXPERIENCE

Founding Full-Stack Engineer

Oct 2025 – Present

Star4ce - dealership retention platform

Irvine, CA (Hybrid)

- Shipped Star4ce, a multi-tenant dealer retention platform (**Next.js/TypeScript, Flask, PostgreSQL**): 44 tables for employees, surveys, ATS, turnover, and billing, with store-level isolation and corporate portfolio access
- Built 250+ **REST** endpoints across 26 **Flask** blueprints with **JWT** + a 5-role / 50+ permission **RBAC** matrix (per-user overrides) so one store cannot read another's employees, surveys, or Stripe billing
- Modeled hiring, 90-day success plans, survey scoring, interview scorecards, and subscription state in **Postgres** and evolved the live schema with idempotent migrations across **Railway** dev/prod
- Built the dealer hub (90+ **Next.js** routes) for 90-day check-ins, survey dispatch, scorecards, turnover analytics, and LLM-backed candidate / command-center reports, backed by 80+ pytest files (400+ tests)

Software Engineer Intern

Feb 2025 – Apr 2025

AcuFlow (Precision Flow Technologies) – pump quoting

Irvine, CA

- After a month of quoting, invoices, and purchases in **QuickBooks**, replaced spreadsheet quoting with a live selector on acuflow.com: sales enter ~30 flow/pressure/material/accessory fields; **Flask** matches a **MySQL** pump catalog (~50 spec columns) using manufacturer rules and picks the closest model at $\leq 2\times$ requested flow
- Generated branded PDF quotes (**ReportLab**, sequential **AQQ IDs**) and emailed the customer plus quotes@acuflow.com so a quote left the site as a numbered artifact, not a spreadsheet export
- Shipped the **API** on **Heroku** behind a GoDaddy/WordPress form (**PHP** proxy); cut quote turnaround from days of manual catalog lookup to a same-session selector used in production at /auto-pump-selector/

EDUCATION

San José State University

Expected May 2028

Master of Science in Artificial Intelligence

San José, CA

- Autonomous Systems Specialization

California State University, Fullerton

May 2024

Bachelor of Science in Computer Science

Fullerton, CA

- Dean's List – Cum Laude, GPA 3.61

PROJECTS

RoadAware | *Python, PyTorch, TorchVision, Ultralytics YOLOv8, OpenCV*

[GitHub](#)

- Built a road-scene object detection pipeline on **BDD100K**, preprocessing **10,000** dashcam images into train/validation/test splits across **8 traffic classes**
- Fine-tuned and compared **YOLOv8m** and **Faster R-CNN ResNet-50 FPN** using augmentation, resolution scaling, and repeat-factor sampling for class imbalance
- Evaluated models with **mAP**, precision, recall, F1, per-class AP, confusion matrices, and latency; best YOLOv8m reached **0.515 mAP@0.5**, **0.734 F1**, and **64.9 FPS**

Gap Reconstruction for Autonomous Vision | *Python, OpenCV, PyTorch, LaMa, YOLO*

[GitHub](#)

- Built a calibrated multi-camera panorama pipeline using **Waymo** images for autonomous driving perception
- Simulated missing field-of-view gaps using camera-ID maps and seam-based masking
- CMPE 297 team project (stitching/integration): evaluated **LaMa** inpainting with **YOLO**; inpainting did not restore detections in masked FOV gaps

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, TypeScript

Frameworks: Next.js, React, Flask, Tailwind CSS

Databases: PostgreSQL, MySQL

Cloud/Tools: Railway, Heroku, Stripe, Git, GitHub, Linux

Machine Learning: PyTorch, OpenCV, YOLO, NumPy

Concepts: REST APIs, JWT/RBAC, OOP